

Linear Programming and Learning with Constraints

Chen-Yu Wei

Linear Programming

$$\max_{f, F} F$$

subject to

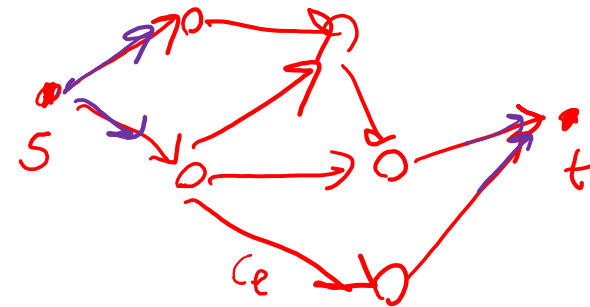
$$\sum_{e \in \text{OUT}(v)} f_e - \sum_{e \in \text{IN}(v)} f_e = \begin{cases} F, & v = s \\ -F, & v = t \\ 0, & v \neq s, t \end{cases}$$

$$0 \leq f_e \leq c_e \quad \forall e \in E$$

source

sink

$\{f_e\}_{e \in E}$



Linear Programing

$$\max_{f,F} F$$

$$\text{s. t.} \quad Bf = Fb$$

$$0 \leq f \leq c$$

Canonical Form of LP

$$\min_x \quad C^T x$$

s.t.

$$a_1^T x \geq b_1$$

$$a_2^T x \geq b_2$$

$$\vdots$$

$$a_N^T x \geq b_N$$

$$x_1 \geq 0$$

$$\vdots$$

$$x_d \geq 0$$

$$\Rightarrow$$

$$Ax \geq b$$

$$x \geq 0$$

Primal and Dual Problems

Primal

$$\alpha = \min_{x \in \mathbb{R}^d} C^T x$$

$$\text{s.t. } Ax \geq b \quad (m \text{ constraints})$$
$$x \geq 0 \quad (d \text{ constraints})$$

$$x \in \mathbb{R}^d \quad (d = \text{dimension of var})$$

$$A \in \mathbb{R}^{m \times d} \quad (m: \# \text{ of constraints})$$

feasible $\rightarrow \infty$

Dual

$$\beta = \max_{y \in \mathbb{R}^m} b^T y$$

$$\text{s.t. } A^T y \leq c \quad (d \text{ constraints})$$
$$y \geq 0 \quad (m \text{ constraints})$$

$$\alpha = \beta$$

$$A^T \in \mathbb{R}^{d \times m} \quad (d: \# \text{ constraints})$$
$$(m = \text{dimension of var})$$

not feasible $\rightarrow \infty$

Weak / Strong Duality Theorem

Theorem. (Weak Duality Theorem)

Primal value $\geq$ Dual value

$$\min \max \geq \max \min$$

$$\alpha \geq \beta$$

Theorem. (Strong Duality Theorem)

If either primal or dual problem is feasible, then

Primal value = Dual value

$$\min \max = \max \min$$

$$\alpha = \beta$$

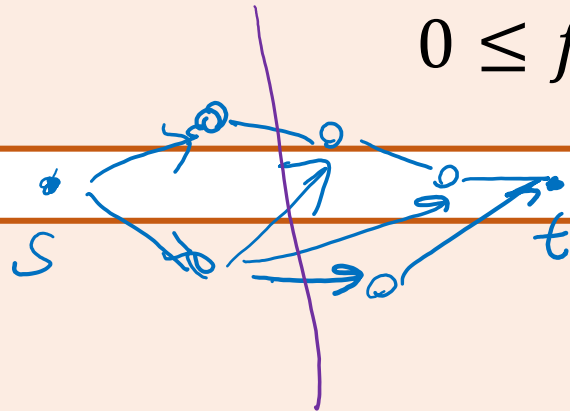
Examples of Primal/Dual Pairs

$$\text{max flow} \leq \text{min cut}$$

Max flow

$$\max_{f, F} F \quad \text{subject to} \quad \sum_{e \in \text{OUT}(v)} f_e - \sum_{e \in \text{IN}(v)} f_e = \begin{cases} F, & v = s \\ -F, & v = t \\ 0, & v \neq s, t \end{cases}$$

$$0 \leq f_e \leq c_e \quad \forall e \in E$$



$\{ S : \text{containing } s \}$
 $\{ T : \text{containing } t \}$

$$\text{value}(\text{cut}) = \sum_{\substack{u \in S \\ v \in T}} c_{uv}$$

min cut

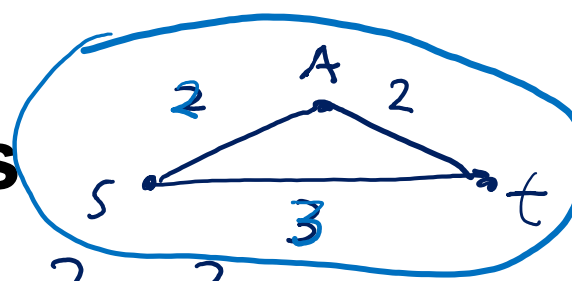
$$\min_{y, z} \sum_{(u, v)} c_{uv} z_{uv} \quad \text{subject to}$$

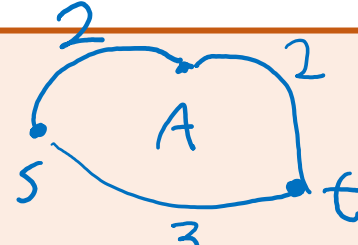
$$z_{uv} \geq y_u - y_v \quad \forall (u, v) \in E$$

$$z_{uv} \geq 0 \quad \forall (u, v) \in E$$

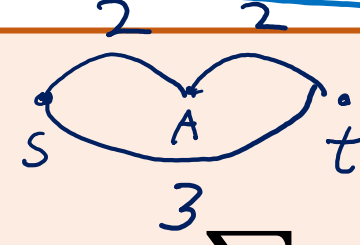
$$y_s - y_t = 1$$

Examples of Primal/Dual Pairs





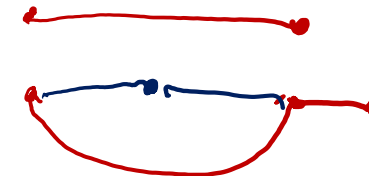
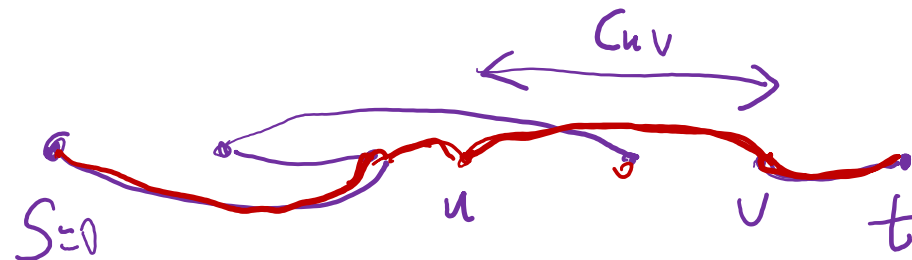
$$\min_{x \geq 0} \sum_{e \in E} c_e x_e \quad \text{subject to}$$



$$\sum_{e \in \text{OUT}(v)} x_e - \sum_{e \in \text{IN}(v)} x_e = \begin{cases} 1, & v = s \\ -1, & v = t \\ 0, & v \neq s, t \end{cases}$$

$$\max_d d_t - d_s \quad \text{subject to} \quad d_v - d_u \leq c_{uv} \quad \forall (u, v) \in E$$

Handwritten notes: $d_s = 0$ (above d_s), d_t (below d_t)



Proof (Idea) of Weak / Strong Duality Theorems

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Solving LP with No-Regret Algorithms

